

Sreejato (Sree) Chatterjee

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RESEARCH INTERESTS

I am interested in biomedical informatics, and the intersection of genomics, machine learning and population health. I am particularly interested in understanding how genetic and epigenetic variation can be studied alongside social and environmental factors to better characterize disease risk. My aim is to develop robust, interpretable, and data-driven methods that integrate genomic and large-scale health data to generate actionable insights for precision medicine & health equity.

EDUCATION

University of Rochester

Rochester, NY

BS in Computational Biology (Honors in Research), BA in Data Science

Aug 2022 – May 2026

Thesis: Epigenetic Regulation of Brain Aging GWAS Variants within Transposable Element Sequences

RESEARCH EXPERIENCE

University of Rochester School of Medicine & Dentistry (SMD)

Rochester, NY

Department of Biomedical Genetics (PI: Dr. Hongbo Liu)

Mar 2025 – May 2026

- Developed an integrative genomics framework to analyze regulatory context of brain-aging GWAS variants.
- Integrated RepeatMasker, snATAC-seq, and ENCODE cCRE annotations for variant characterization.
- Built bioinformatics pipelines in Python & R for TE annotation and enrichment analysis.
- Performed pathway mapping of GWAS variants using clusterProfiler and statistical testing.
- Applied partitioned LD Score Regression to evaluate heritability enrichment of TE-overlapping variants.

University of Rochester School of Medicine & Dentistry (SMD)

Rochester, NY

Department of OB/GYN (PI: Dr. Timothy Dye)

Sep 2024 – May 2026

- Developed an LLM-based pipeline to classify structural oppression from free-text identity data.
- Analyzed 11,000+ responses across 170+ countries from a global COVID-19 survey.
- Benchmarked multiple LLMs against expert-labeled data across prompting strategies
- Conducted cross-country error analysis to evaluate model validity and bias.
- Performed multivariable statistical analysis linking AI-derived scores to health and socioeconomic outcomes.

National Institute of Neurological Disorders & Stroke (NINDS)

Bethesda, MD

Summer Research Intern, Wagner Lab, Neo-Cortex Cerebellum Circuitry Unit

Jun 2024 – Aug 2024

- Studied cerebellar learning mechanisms in dopamine self-stimulation tasks in transgenic mice.
- Developed real-time tracking and optogenetic stimulation pipelines for closed-loop CPP experiments.
- Implemented video tracking models using Bonsai and Python.
- Applied DeepLabCut for pose estimation of mouse behavior.
- Built Python algorithms to quantify time spent in reward zones and visualize movement patterns.

National Cancer Institute (NCI)

Bethesda, MD

Summer Research Intern, Vaccine Group, Center for Immuno-Oncology (CIO)

Jun 2023 – Aug 2023

- Investigated combination immunotherapy effects in HPV-related tumor models.
- Measured immune protein levels in tumor microenvironment and peripheral tissues.
- Quantified tumor immune cell populations and anti-tumor response.
- Performed fluorescence assays on serum and tumor samples from murine models.
- Analyzed gene expression using qPCR and cytokine/chemokine profiling.

LEADERSHIP & SERVICE

University of Rochester Out in STEM (oSTEM)

Rochester, NY

Co-President (2025–26), Secretary (2024–25)

Jan 2024 – Present

- Organized pre-professional programming for LGBTQ+ students in STEM, including resume workshops and graduate application sessions.

- Coordinated travel and funding for 8–10 students to attend the oSTEM National Conference.

University of Rochester Alpha Phi Omega, Mu Lambda Chapter

Rochester, NY

Co-Vice-President of Service (2025–26), Secretary (2024–25)

Jan 2024 – Present

- Completed 160+ hours of community service and coordinated 500+ service hours for 30+ members.
- Partnered with 10+ local nonprofits, including House of Mercy and Rochester Greenovation.
- Organized workshops, guest speakers, and a regional service project with Ronald McDonald House.
- Managed chapter logistics, including scheduling, event coordination, and partnerships.
- Awarded Outstanding Student Service Award (2026) for leadership and service impact.

University of Rochester Neurodiversity Alliance

Rochester, NY

Co-Chapter Leader (2023–26)

Sep 2023 – Present

- Led a team of 5+ mentors supporting students with learning disabilities in local schools.
- Conducted 60+ hours of mentorship focused on self-advocacy and metacognitive skills.
- Organized outreach events on neurodiversity awareness and inclusion.

University of Rochester Office of Residential Life & Housing

Rochester, NY

First-Year Fellow

Sep 2023 – May 2025

- Mentored first-year students on academic planning and campus resources.
- Organized academic and wellness events, including faculty panels and study sessions.

University of Rochester Chemistry Department

Rochester, NY

Teaching Assistant

Jan 2024 – May 2024

- Led weekly workshops and office hours for CHEM 132 (quantum mechanics, thermodynamics, kinetics).
- Supported grading and facilitated problem-solving sessions for undergraduate students.

HONORS & AWARDS

- **Charles L. Newton Prize, UofR Hajim School of Engineering 2026**
 - Recognizes engineering students who show special proficiency in an engineering subject and have conducted research, given a presentation, or published a paper of special merit.
- **Region Q Outstanding Student Service Award, Alpha Phi Omega 2026**
 - Recognizes active fraternity brothers within Region Q who have made service a priority by performing an extraordinary amount of service hours or planning a unique service project.
- **Research Presentation Award, UofR Office of Undergraduate Research 2026**
 - Financial award of \$1,500 for presentation of academic research at the National Conference on Undergraduate Research (NCUR).
- **Medallion Leadership Society Inductee, University of Rochester 2025–2026**
 - Recognized for completing a multi-stage leadership development program and executing a Neurodiversity Awareness Fair to increase campus awareness and inclusion.
- **First Place in Biomedical Data Science Hackathon Fall 2025 (Undergraduate Division)**
 - Developed a gene expression prediction model for combinatorial perturbations, achieving top overall performance among competing teams.
- **Neurodiversity Alliance Scholarship Recipient 2025**
 - National award supporting high-achieving neurodivergent students (\$2,500/year).
- **Second Place in RAISE AI & Informatics Competition (Undergraduate Division) 2025**
 - Led an NLP-driven analysis of 14,000+ news headlines to characterize societal impacts of AI, advancing to finals and placing 2nd out of 119 teams.
- **First Place Gold Award in Best Visualization Category, RIT ASA Datafest 2025**
 - Produced high-impact data visualizations and predictive insights from commercial real estate datasets, earning the Gold Medal award from industry judges.
- **First Place in Biomedical Data Science Hackathon Fall 2024 (Undergraduate Division)**
 - Built a machine learning model for gene-regulatory associations, achieving top undergraduate perfor-

mance and ranking among leading overall teams.

- **Out 4 Undergrad Life Sciences Conference Scholar 2024**

- Selected to attend an all-expenses-paid, highly competitive national conference supporting LGBTQ+ students in life sciences and research careers.

- **Dean's Scholarship Recipient 2022–2026**

- Awarded a merit-based scholarship for academic excellence and leadership (\$20,000 annually).

PUBLICATIONS

1. **S. Chatterjee et al.** “Assessing Historical Structural Oppression Worldwide via Rule-Guided Prompting of Large Language Models,” in *2025 IEEE International Conference on Big Data (BigData)*, 2025, pp. 2783–2792, doi: 10.1109/BigData66926.2025.11401510.

ORAL PRESENTATIONS

1. **S. Chatterjee.** “Measuring Global Structural Oppression in Free-Text Identity Data via Rule-Guided Large Language Models.” In *UR Undergraduate Research Expo Social Sciences Speakers Symposium*, Rochester, NY, 2026.
2. **S. Chatterjee.** “Measuring Global Structural Oppression in Free-Text Identity Data via Rule-Guided Large Language Models.” In *oSTEM National Conference*, Baltimore, MD, 2025.

POSTER PRESENTATIONS

1. **S. Chatterjee et al..** “Epigenetic Regulation of Brain Aging GWAS Variants within Transposable Element Sequences.” In *National Conference on Undergraduate Research (NCUR)*, Richmond, VA, 2026.
2. **S. Chatterjee.** “Measuring Global Structural Oppression in Free-Text Identity Data via Rule-Guided Large Language Models.” In *7th National Big Data Health Science Conference*, Columbia, SC, 2026.
3. **S. Chatterjee.** “Integrative Genomic Analysis Reveals Function of GWAS Variants Within Transposable Elements in Human Diseases.” In *URMC Aging Research Day Conference*, Rochester, NY, 2025.
4. **S. Chatterjee.** “Measuring Global Structural Oppression in Free-Text Identity Data via Rule-Guided Large Language Models.” In *oSTEM National Conference*, Baltimore, MD, 2025.
5. **S. Chatterjee.** Additional poster presentations at *Finger Lakes Science and Technology Showcase* (2025, 2026), *UR Undergraduate Research Poster Expo* (2024, 2026), *NIH Summer Poster Day* (2023, 2024), and *NCI Center for Cancer Research Poster Day* (2023).

TECHNICAL SKILLS

- **Programming:** Python, R, Bash, SQL, Java
- **AI/ML:** PyTorch, TensorFlow, scikit-learn, NumPy, Pandas, Statsmodels
- **NLP & Text Mining:** Hugging Face Transformers, LLMs, spaCy, BERTopic, VADER, TextBlob
- **Bioinformatics:** Seurat, ScanPy, Bowtie2, PLINK, LDSC, BEDTools, RepeatMasker, GWAS, ENCODE
- **Imaging & Behavior:** DeepLabCut, Bonsai, ImageJ
- **Tools & Systems:** Git, Linux, Conda, Jupyter
- **Lab Techniques:** PCR, qPCR, Gel Electrophoresis, Western Blotting, Cell Culture, Flow Cytometry, DNA Extraction, Immunohistochemistry, Tissue Sectioning, ELISA, Fluorescence Microscopy